

Title Page

Pan-Andean explainable machine learning identifies priority glacial lakes for outburst flood early warning

Authors

Richar Andre Vilca-Solorzano^{1,*}

ORCID: 0009-0003-2385-5263

Dina Maribel Yana-Yucra¹

ORCID: 0009-0003-6218-2735

Milton Vladimir Mamani-Calisaya¹

ORCID: 0000-0000-0000-0000 (*pending*)

Fred Torres-Cruz¹

ORCID: 0000-0003-0834-6834

Affiliation

¹ Faculty of Statistical and Computer Engineering, Universidad Nacional del Altiplano (UNAP), Puno, Peru.

Corresponding author

Richar Andre Vilca-Solorzano

Faculty of Statistical and Computer Engineering,

Universidad Nacional del Altiplano (UNAP), Puno, Peru

E-mail: 75521963@est.unap.edu.pe

Statements and Declarations

Funding. The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Competing Interests. The authors have no relevant financial or non-financial interests to disclose.

Ethics approval. Not applicable.

Consent to participate / publish. Not applicable.

Author Contributions. Richar Andre Vilca-Solorzano conceptualized the study, developed the methodology, performed the formal analysis and visualization, and wrote the original draft. Dina Maribel Yana-Yucra contributed to data curation, validation, and writing – review & editing. Milton Vladimir Mamani-Calisaya and Fred Torres-Cruz contributed to supervision, resources, and critical review of the manuscript. All authors read and approved the final manuscript.

Data Availability. Inventory, models, SHAP values, and the reproducible pipeline are openly available at <https://github.com/Andre031222/pan-andean-glof-susceptibility> (Zenodo DOI on acceptance).

Note: replace the placeholder ORCID iDs (0000-0000-0000-0000) with the real identifiers before submission.